

FRANCOIS SANDY LEUTOU

Providence, RI | (401) 678-7241 | francois_sandy_leutou@brown.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

Brown University — Providence, RI

Expected May 2029

Sc.B Applied Math-Computer Science

- **Relevant Coursework:** CSCI0190: Accelerated Intro to CS, CSCI0300: Fundamentals of Computer Systems, CSCI1970KF: Agentic Studio, APMA1655: Intro to Probability & Statistics, MATH0520: Linear Algebra

GBHS Nkol-Eton — Yaounde, Cameroon

June 2023

- GCE A-Levels (4.0 GPA): Computer Science, Mathematics, Physics, Chemistry, Biology
- GCE O-Levels (4.0 GPA): Computer Science, Mathematics, Physics, Chemistry, Biology, Economics, Geography, English, French, Additional Mathematics

PROJECTS

Trackable — Full-Stack Productivity Platform for College Students

| React, TypeScript, Python, FastAPI | [GitHub](#) | May 2026 - June 2026

- Built a full-stack web app that unifies assignments, deadlines, events, and tasks from Canvas, Google Calendar, Gradescope, and ICS course feeds into a single responsive timeline, with smart reminders and study tips.
- Engineered a FastAPI/SQLAlchemy backend with JWT authentication, encrypted credential storage, rate limiting, SSRF protection, and a fail-closed production config, plus a Vite + React + TypeScript frontend using TanStack Query.
- Designed a pluggable integration layer with a demo-data mode, letting the app run end-to-end without external API keys; containerized with Docker and deployed via Railway (backend) and Vercel (frontend) behind a GitHub Actions CI pipeline.
- Wrote 75+ automated tests (Pytest + Vitest) and leveraged AI-assisted development (Claude Code and codex) to accelerate design, implementation, and testing.

Document Search & Tagging System | Python, FastAPI, React, sentence-transformers | [GitHub](#) | [Live Site](#) |

May 2026

- Built a hybrid NLP information retrieval system over 75+ policy and research documents, combining sentence-transformers (all-MiniLM-L6-v2) semantic embeddings with TF-IDF scoring to surface the most relevant documents from natural language queries with sub-200ms latency via FAISS vector indexing
- Applied unsupervised k-means clustering over document embeddings to auto-generate topic tags across 8 domain clusters without labeled training data, achieving coherent groupings validated against ground-truth topic labels
- Offloaded pre-computed model artifacts (embeddings, FAISS index) to Cloudflare R2 object storage, fetched at backend startup on Railway — eliminating cold-start recomputation and keeping the server image lightweight
- Configured GitHub Actions CI pipeline (backend-pytest; + frontend-vitest; test suites) with branch protection on main (required passing CI, up-to-date branch, PR-only merges) and auto-deploy to Railway/Vercel on merge, enforcing a production-grade release flow
- Designed and deployed a FastAPI backend exposing RESTful search and tagging endpoints, integrated with a React frontend for real-time interactive querying

LeetCode Progress Tracker (Full-Stack Web App) | Python, React, SQLAlchemy, FastAPI | [GitHub](#) | [Live Site](#) | May 2026

- Designed a normalized relational database schema (PostgreSQL) with efficient indexing to support sub-second filtering and analytics across problem metadata (difficulty, topic, project title), reducing query overhead through deliberate schema design
- Built a RESTful FastAPI backend implementing full CRUD operations with SQLAlchemy ORM; modeled entity relationships to support future multi-user isolation at the data layer
- Deployed a production full-stack architecture across the complete SDLC — from local development to staged testing to live deployment — with Vercel-hosted React frontend, Railway-hosted FastAPI backend, and managed PostgreSQL instance; enforced branch protection and CI gating
- Leveraged AI-assisted development tools (Claude Code, GitHub Copilot) to accelerate frontend prototyping and feature iteration

Image Classification Web App | Python, React, FastAPI, PyTorch | [Github](#) | [Live Site](#) | April 2026 - May 2026

- Deployed a deep learning inference service using a pretrained ResNet50 CNN (PyTorch) via a FastAPI backend, serving real-time image classification through a RESTful API with model-ready preprocessing (resize, normalize, tensor conversion)
- Architected a client-server model deployment pipeline — from trained model weights to live HTTP endpoint — enabling scalable inference without reloading the model between requests
- Designed a React frontend surfacing top-3 predictions with confidence scores, reducing average user time-to-result to under 2 seconds end-to-end

EXPERIENCE

Software Engineering Intern | Amea | Yaounde, CMR (Remote) | Python, PyTorch, NumPy, HuggingFace | May 2026 – Aug 2026

- Built a full data retrieval, pre-processing, and storage system to collect historical games for a specific player and for players within a certain elo range for use in training chess twin models.
- Designed and built a training pipeline in python using PyTorch to facilitate easy development of chess twins for various people and elo ranges with a simple config extraction system and a generalized training schema.
- Designed encoding pipeline to safely provide good quality data for use by the model in processing, with a (B, 4184) resulting numpy array from a 17x8x8 board.
- Created multi-layered prediction model, working on roughly 9.5 million parameters to achieve maximized precision and accuracy on player move prediction.
- Pretrained an initial trunk on 1million extracted Lichess positions from the HuggingFace Lichess monthly dump and finetuned on 100000 player positions, visualizing progress using a TensorBoard, effectively achieving <1.9 masked cross entropy loss and a top1 move accuracy of 44% matching SOTA model maia within 6% accuracy, with 67% top3 and 77% top5 accuracy.
- Built prediction pipeline with move encoding, prediction, and random move selection within top3 move with reference given to top1 to mimic human variability.

Software Engineering Intern | Amea | Yaounde, CMR (Hybrid) | *May 2025 – August 2025*

- Collaborate cross-functionally on development, release, and maintenance of production software for a growing startup, participating in weekly sprint planning and progress reviews

Program Facilitator & SAT Tutor | Open Dreams | Yaounde, CMR | *Sep 2024 – Aug 2025*

- Facilitated SAT prep sessions for 30+ students, driving score improvements of 100+ points per student through targeted instruction and personalized feedback
- Strengthened public speaking and instructional design skills by leading structured group sessions

Co-Founder & Head Administrator | Grade Masters Tutoring Community | Yaounde, CMR | *Jan 2024 – May 2025*

- Founded and scaled tutoring community to 20+ students; organized mentor-tutor pairings and directly mentored 2 students, helping 4 achieve 3.5+ GPAs
- Taught Math and Computer Science (~10 hrs/week), designing curriculum to target individual student weaknesses

CS Teaching Assistant (A & O-Level) | GBHS Nkol-Eton | Yaounde, CMR | *Oct 2021 – Jun 2023*

- Led revision sessions and supervised programming labs for A/O-Level CS students, contributing to 100% A-Level pass rate across cohort
- Assisted with grading, administrative work, and curriculum support (~5 hrs/week)

CS Club President | GBHS Nkol-Eton | Yaounde, CMR | *Oct 2022 – Apr 2023*

- Led 13-member club; taught C programming fundamentals to juniors and mentored peers through end-of-year CS projects, strengthening team's technical foundations

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks: React, FastAPI, PyTorch, Flask, Tailwind CSS, SQLAlchemy

ML / AI: Deep Learning, NLP, Information Retrieval, Sentence Transformers, TF-IDF, FAISS, scikit-learn (k-means clustering), ResNet / CNNs

Databases: PostgreSQL, Relational Schema Design, SQLAlchemy ORM

Cloud & Infra: Cloudflare R2, Vercel, Railway

CI/CD: GitHub Actions, Branch Protection, Auto-deploy (Vercel / Railway)

Developer Tools: Git, Docker, Bash, Linux, AI-assisted tools (Claude Code, GitHub Copilot)

Human Languages: French (Native), English (Fluent), Japanese (Beginner)

Interests: Hiking, psychology, soccer, chess, reading, pool